

Math 142, F19
Quiz 12

Name: *Answer*

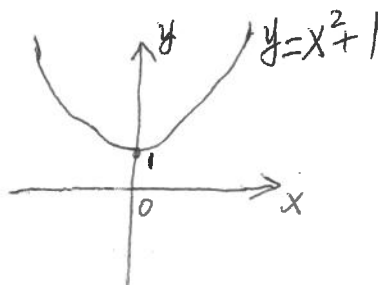
Instructions: You must show all of your work to receive credits for correct answers.

Problem 1. (10 points) Find the corresponding Cartesian equations for the following parametric equations:

(a) $x = 3t$, $y = 9t^2 + 1$, $-\infty < t < \infty$.

$$y = (3t)^2 + 1$$

$$\Rightarrow \underline{y = x^2 + 1}$$

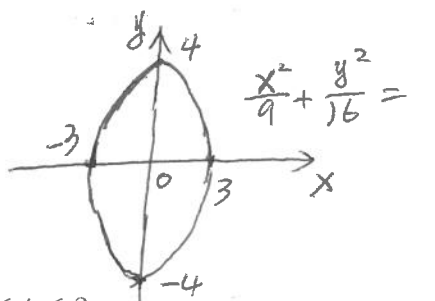


(b) $x = 3 \cos t$, $y = 4 \sin t$, $0 \leq t \leq 2\pi$.

$$\frac{x}{3} = \cos t, \frac{y}{4} = \sin t \Rightarrow \left(\frac{x}{3}\right)^2 + \left(\frac{y}{4}\right)^2 = \cos^2 t + \sin^2 t = 1$$

$$\Rightarrow \underline{\frac{x^2}{9} + \frac{y^2}{16} = 1}$$

$\downarrow$ $\downarrow$
 3^2 4^2



(c) $x = 1 + 2 \cos t$, $y = 2 \sin t - 2$, $0 \leq t \leq 2\pi$.

$$x - 1 = 2 \cos t, y + 2 = 2 \sin t$$

$$\Rightarrow (x-1)^2 + (y+2)^2 = (2 \cos t)^2 + (2 \sin t)^2$$

$$\underline{(x-1)^2 + (y+2)^2 = 4 \rightarrow 2^2}$$

